

Group-Fair Stochastic Gradient Descent via Masked Optimal Transport

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Table 1: stub

Lemma 1. *stub*

Lemma 2. *stub*

Theorem 3. *stub*

Theorem 4. *stub*

Corollary 5. *stub*

Theorem 6. *stub*

(1)

(2)

Proposition 7. *stub*

A APPENDIX

A. Proofs. *Proposition 7.* Conditionally on $A^t = A_j$ the within-group averages are unbiased for ∂f_i , so $\mathbb{E}[\text{batch subgradient} \mid A^t = A_j] = \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] \partial f_i(\theta)$; averaging over q gives $\sum_i \tilde{p}_i \partial f_i = \partial F_{\tilde{p}}$, and $\sum_i \tilde{p}_i = \sum_j q_j \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] = 1$. Both bounds follow from $0 \leq f_i \leq M$ and the two-point inequality $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq [F_p - F_{\tilde{p}}](\theta_{\tilde{p}}^*) + [F_{\tilde{p}} - F_p](\theta^*)$.

Sinkhorn scaling. Starting from any $T^{(0)} > 0$ on E with $\mathbf{1}^\top T^{(0)} = q$, alternate

$$T \leftarrow \text{diag}(p \oslash T \mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (3)$$

until $\|T \mathbf{1} - p\|_1 \leq \varepsilon$. Under (1) the iteration converges to a solution of (??) [?, ?], being block-coordinate ascent on the dual of the entropic problem $\min\{\sum_E T \log T : T \mathbf{1} = p, \mathbf{1}^\top T = q\}$.

Lemma 1. Conditionally on $A^t = A_j$, $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$ for $i \in A_j$. Averaging over $A^t \sim q$, $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$, the row-marginal constraint being used in the last step.

Lemma 2. $Y[\cdot, j]$ is a probability vector on A_j , so $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$ by convexity of $\|\cdot\|^2$; take expectations and use $\sum_j T[i, j] = p_i$: $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$.

Theorem 6 (sketch). Route supply p_i from a source through group i , along the edges of E with infinite capacity, into composition j with demand q_j , and to a sink. A coupling exists iff the maximum flow saturates every source edge, which by max-flow/min-cut [?] is the right inequality of (1); the left one is the same statement for the reversed network. On singletons (1) is Hall’s condition [?].

Corollary 5. For $I = \{i\}$, $\mathcal{N}(I)$ is the set of compositions containing i , of total mass r_i , and $\mathcal{S}(I) = \{j : A_j = \{i\}\}$. Substitute $r_i = 1 - (1 - \pi_i)^b$ and $r_i = K/N$.

Theorem 3. By non-expansiveness of Π_C , $\|\theta^t - \theta^*\|^2 \leq \|\theta^{t-1} - \theta^*\|^2 - 2\eta \langle \hat{g}^t, \theta^{t-1} - \theta^* \rangle + \eta^2 \|\hat{g}^t\|^2$. Conditioning on θ^{t-1} , Lemma 1 and convexity bound the inner product below by $F_p(\theta^{t-1}) - F_p(\theta^*)$, and Lemma 2 bounds the last term by $\eta^2 G^2$. Summing, telescoping, and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, and the stated η balances the two terms [?]. In full: $\mathbb{E}\|\theta^t - \theta^*\|^2 \leq \mathbb{E}\|\theta^{t-1} - \theta^*\|^2 - 2\eta \mathbb{E}[F_p(\theta^{t-1}) - F_p(\theta^*)] + \eta^2 G^2$; summing over $t \leq T$, telescoping, dropping the terminal distance and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, which is $DG/\sqrt{T}$ at $\eta = D/(G\sqrt{T})$; the same bound holds for $\eta_t = \eta_0/\sqrt{t}$ up to constants.

Theorem 4. Add and subtract $F_p(\theta^\dagger) = \Phi(\tilde{p})$; the floor is exact by definition and $\Phi(p) = F_p(\theta^*)$. Theorem 3 applied to $F_{\tilde{p}}$ gives the residual bound on the surrogate; the residual on F_p differs from it by $(F_p - F_{\tilde{p}})(\bar{\theta}_T) - (F_p - F_{\tilde{p}})(\theta^\dagger)$, which vanishes as $\bar{\theta}_T \rightarrow \theta^\dagger$ (for instance under strong convexity, which both of our instances satisfy). The ℓ_1 bound is the two-point argument of Proposition 7 with w for $\tilde{p}$.

B. The infeasible regime in closed form (IMDb-Wiki, $\beta = 3$). On the mirrored instance, where 41 of 100 groups violate $p_i \leq r_i$ and hold 88% of the importance mass, the Sinkhorn limit truncates the target at the observation-rate ceiling, $\hat{p}_i \approx \min(p_i, r_i)$ up to renormalization (maximum

deviation 0.023), giving $\|p - \hat{p}\|_1 = 1.58$ of a maximum possible 2. Because the task is linear regression, the minimizers θ^* of F_p and $\theta^\dagger$ of $F_{\tilde{p}}$ are available in closed form: $\Phi(p) = F_p(\theta^*) = 82.9$ (matching the measured full-coverage plateau, 82.95) and $\Phi(\tilde{p}) = F_{\tilde{p}}(\theta^\dagger) = 105.9$. The alignment floor is therefore 23.0, and the realized bias equals its exact expression $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$, where $L(\theta)$ collects the per-group risks. The ℓ_1 form $2M\|p - \hat{p}\|_1$ of Theorem 4 holds but is conservative here, since any uniform bound satisfies $M \geq \max_i L_i(\theta^\dagger) \approx 454$. The group-blind surrogate $\tilde{p}_i = r_i/K$ has the same floor to two decimals ($\Phi(\tilde{p}) = 105.9$); both surrogates starve the same groups, which is why Δ_{floor} vanishes at this endpoint of Table 1.

E. Instances and protocol, in full. Adult: logistic regression with $d=109$ features including an intercept, cross-entropy loss; 100 groups of 30 samples, each homogeneous in race, with 85/10/3/1/1 groups for White/Black/Asian-Pac./Amer-Ind./Other; the target gives each race 1/5, split equally among its groups, so on the prevalence instance the five groups of the three smallest races hold 60% of the target against $r_i \approx K/m = .03$. IMDb-Wiki: the first 100 identities with 30 images, 128-d ResNet embeddings, linear regression of age; at $\beta=0$ nine groups are infeasible (mass .31), at $\beta=3$ forty-one (.88). Each step draws $K=3$ groups without replacement with probabilities $\propto r$, inducing q over the $\binom{100}{3}$ compositions; q is estimated from 10^6 draws and the plan fitted once per instance by (3), which on every instance with $\nu = 0$ reaches a row error below $3 \cdot 10^{-8}$, certifying the remaining inequalities of (1). All rules consume the same $H=5$ local updates from the same draw A^t ; the fixed multiplier is $(m/K)p_i$, the rescaling that presumes uniform observation, and full is the reference $\Phi(p)$. $\arg \min F_w$ is closed-form for least squares and Newton to machine precision for the logistic loss.

C. Adult severity sweep. The five Adult skews omitted from Table 1, same conventions.

β	ν	Δ_{floor}	gain	t
0	.60	.0110	+.0210	44.7
0.25	.60	.0144	+.0181	69.4
0.5	.40	.0129	+.0134	35.2
0.75	0	.0049	+.0050	35.8
1	0	.0003	+.0002	7.1

D. Adult per-race cross-entropy, prevalence regime. Tail-500 over 5 seeds, decaying stepsize.

	White	Black	Asian-Pac.	Amer-Ind.	Other
GAVOT	.339	.179	.368	.175	.073
group-blind avg.	.332	.200	.399	.189	.119
full	.379	.177	.328	.123	.031

Transport improves four of five categories and regresses only the majority group, the expected effect of reweighting toward a category-uniform target; the residual loss on the two smallest categories is the unreachable mass of Cor. 5.